Moule Lin

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EDUCATION

Trinity College Dublin QS Top100 2024.04 - 2027.06
Computer Science Doctor Computer Science and Statistics Ph.D. Candidate, Dublin, Ireland
MSCA, **Marie Skłodowska-Curie Fellow**

Eötvös Loránd University(U.S.News #368) 2021.09 - 2023.06
Artificial Intelligence Master Faculty of Informatics Master, Budapest, Hungary
GPA: 3.4/4

- **Stipendium Hungaricum** Scholarship (Fully Scholarship)
- **Excellent thesis**
- **SH Network Mentor**

Northeast Forestry University 211 2020.09 - 2024.06
Computer Science Master College of Computer and Control Engineering Master, Harbin, China
GPA : 4.38/5
Graduate National Scholarship 2022,
Three times First Class Scholarship for Postgraduate

Tianjin Agricultural University 2015.09 - 2019.06
Software Engineering Bachelor School of Information and Computer Engineering Bachelor, Tianjin, China
GPA: **3.36/4**

- Algorithm Competition**, National,** Blue Bridge Cup **: ranking 27/51282 (National First Prize)**
- Algorithm Competition**, Regional,** Blue Bridge Cup **: Tianjin First Prize**
- **People's Government** Scholarship, Tianjin, China
- First Class Academic Scholarship, China
- **Honorary Graduate, Bachelor**

PUBLISHED PAPERS

[1] **Lin, M**., Guan, S., Patane, A., Gregg, D., & Botterweck, G. Bayesian-LoRA: Probabilistic Low-Rank Adaptation of Large Language Models (ICML'26)

[1] **Lin, M**., Patane, A., Jing, W., Guan, S., & Botterweck, G. (2026). Flow-Induced Diagonal Gaussian Processes. The Association for the Advancement of Artificial Intelligence (AAAI'26)

[2] **Lin, M**., Guan, S., Jing, W., Botterweck, G., & Patane, A. Stochastic Weight Sharing for Bayesian Neural Networks. In International Conference on Artificial Intelligence and Statistics **(**AISTATS'25**)**

[3] **Lin, M**., Jing, W., Zou, W., Qiu, Z., & Li, C. . Gaussian-Based Swap Operator for Context-Aware Extraction of Building Boundary Vectors. IEEE Transactions on Geoscience and Remote Sensing. (TGRS)

[5] Guan, S., **Lin, M**., Xu, C., Zhao, J., & Greene, D. Teaching VLMs to Admit Uncertainty in OCR from Lossy Visual Inputs (ICLR'26)

[4] Guan, S., **Lin, M**., Xu, C., Liu, X., Zhao, J., Fan, J., Xu, Q., & Greene, D. PreP-OCR: A complete pipeline for document image restoration and enhanced OCR accuracy. In Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics **(**ACL'25, main**)**

[5] Guan, S., Xu, C., **Lin, M**., & Greene, D. Effective synthetic data and test-time adaptation for OCR correction. In Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing **(**EMNLP'25, main)

Under review:

[1] **Moule Lin**, Qizhen Lan, David Gregg, Goetz Botterweck. Decoupled On-Policy Distillation for LLM Reasoning **(NeurIPS'26)**

[2]**Moule Lin**, Qizhen Lan, David Gregg, Goetz Botterweck. Self-Diagnostics and Rich Execution Feedback to Distill Models for Specific Software Engineering Tasks **(ASE'26)**

INTERNSHIP EXPERIENCE

Ericsson (Budapest) 2021.07 - 2021.10
Researcher Model fusion Budapest

- Specific responsibilities: part of **CUDA** programming, some new algorithms that need to be implemented through **CUDA/C++**.
- For example, Shift's CUDA operator, implemented and applied it to one of the small tasks to accelerate **inference.**
- Multi-modal data, including video, audio, text**,** etc., Transformer model - modified to a local Transformer to focus on key information, like **Swin Transformer.**
- Analyze the data in the communication network to predict user behavior, needs, and preferences, help communication operators optimize services, improve user experience, etc.

Geely Group - Zeekr 2023.07 - 2023.11
Researcher in Autonomous Driving ShangHai

- Researched SOTA autonomous driving and BEV models for automatic parking using fisheye cameras and LiDAR.
- Improved large-scale training efficiency (~8%) by prefetch data loading on 2M+ images/videos.
- Developed a low-light and dirty-camera detection model from data collection to deployment.
- Deployed models with ONNX and TensorRT on the Zeekr vehicle platform

Ubotica — Marie Skłodowska-Curie Secondary Institution
Researcher

- Deploy uncertainty-aware models on satellites for reliable onboard prediction and decision-making
- Apply my previous uncertainty estimation and Bayesian learning research to Ubotica's existing models

PROJECT

Marie Skłodowska-Curie Fellow 2024.05 - Present
Trinity College Dublin Computer Science Dublin, Ireland
Marie Skłodowska-Curie Fellow - LLMs uncertainty estimation

Master Collaboration&Innovation Summer Project - Model interpretability research - pruning, distillation and sparse model research - Eötvös Loránd University 2022.07 - 2022.10
Researcher AI Lab - Eötvös Loránd University Budapest

- Conducted research on model compression, including pruning, distillation, and sparse architectures.
- Proposed a novel "Swap" structure achieving ~10% inference acceleration over the baseline.
- Co-authored the paper Optimized Vectorizing of Building Structures with Swap.
- Maintained CUDA/C++ tutorials and related development materials.